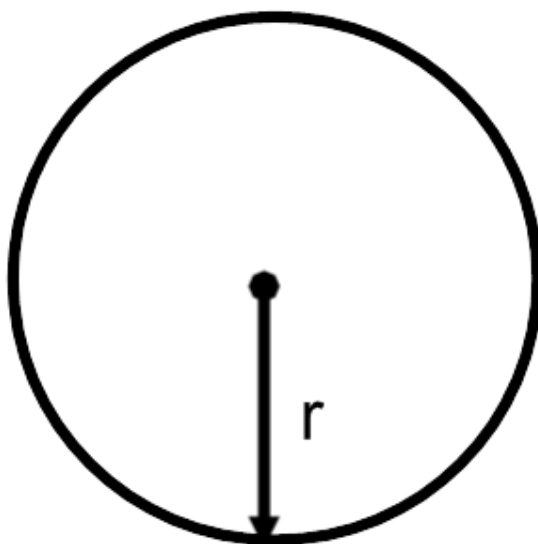


Problem 1

Problem 1

Problem 1 *The radius of a circle is increasing at a rate of 2 inches per minute.*



- (a) *At what rate is the circumference of the circle changing when the radius is 10 inches?*

Explanation. We know: $\frac{dr}{dt} = 2$ inches per minute and we want to find $\frac{dc}{dt}$ when $r = 10$.

Problem 1

$$\begin{aligned}c &= 2\pi r \\ \frac{dc}{dt} &= 2\pi \frac{dr}{dt} \\ \left[\frac{dc}{dt} \right]_{r=10} &= 2\pi(2) \\ \left[\frac{dc}{dt} \right]_{r=10} &= 4\pi \text{ inches per minute}\end{aligned}$$

(b) *At what rate is the area of the circle changing when the radius is 12 inches?*

Explanation. We know: $\frac{dr}{dt} = 2$ inches per minute and we want to find $\frac{dA}{dt}$ when $r = 12$.

$$\begin{aligned}A &= \pi r^2 \\ \frac{dA}{dt} &= 2\pi r \frac{dr}{dt} \\ \left[\frac{dA}{dt} \right]_{r=12} &= 2\pi(12)(2) \\ \left[\frac{dA}{dt} \right]_{r=12} &= 48\pi \text{ inches squared per minute}\end{aligned}$$